

Technical Document #620: Data Engineering - Comprehensive Overview

Technical Document #620: Data Engineering - Comprehensive Overview

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Data governance establishes policies and procedures for managing data assets.
- Data lineage tracks the origin and movement of data through systems.
- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a data warehouse.